

The *GPR143* gene is located on the X chromosome and mutations in this gene can cause ocular albinism (OA), a recessive condition in which pigmentation in the eye is affected. A couple has three children: two of them (a girl and a boy) have OA but a third child (a girl) is not affected. Which of the following conclusions about the inheritance of OA in this family can be drawn?

- ☐ A. The parents are both OA carriers. [28%]
- ☐ B. The disorder will likely be passed to 100% of their sons. [14%]
- ☐ C. The disorder will likely be passed to 25% of their daughters. [7%]
- ✓ ☒ D. The father has OA. [49%]

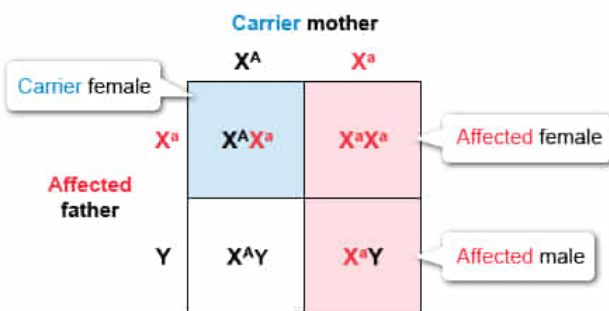
Correct

50% answered correctly

Explanation:

X-linked recessive inheritance of ocular albinism

X^A = Wild-type *GPR143* allele X^a = Mutant *GPR143* allele



- Males cannot be **carriers** of X-linked recessive alleles; if they have the allele, they will be **affected**.
- Female **carriers** of X-linked recessive alleles will pass the allele to 50% of their sons.

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Human somatic cells contain 22 pairs of **autosomes** and two **sex chromosomes** (ie, X and Y). However, unlike autosomes sex chromosomes are not fully **homologous** and most **genes** on the X chromosome do not have a counterpart on the Y chromosome, leading to different inheritance patterns for X-linked genes.

In **X-linked inheritance**, females receive two copies of each gene on the X chromosome, but males receive only one copy. For males, the terms **homozygous** or **heterozygous** do not apply to the X chromosome and, accordingly, males cannot be **carriers** of a **recessive** trait. Therefore, if a male inherits a recessive **allele** from the X chromosome, the trait will always be

expressed.

The question asks what conclusions can be drawn about this disorder's inheritance, given that it is X-linked recessive, and the couple has both a son and daughter with OA and a daughter without OA. For this situation to occur,

the **father must have OA** (X^aY) and the mother must be a carrier (X^AX^a). Even though the father possesses only one X^a allele, he will have the condition because he does not have a second X chromosome.

(Choice A) It is not possible for males to be carriers of X-linked genes, because males cannot be heterozygous for X-linked traits. Only females can be carriers of X-linked genes; therefore, both parents are *not* carriers.

(Choice B) A 50% (*not* 100%) probability exists that sons of this couple will be affected. Regardless of the father's status, 100% of sons would be affected only if the mother also had OA (X^aX^a) because only mothers can pass X chromosomes to their sons.

(Choice C) The probability that daughters of this couple will be carriers is 50%, *not* 25%. In autosomal recessive inheritance, if each parent has one recessive allele, the probability that their child will inherit both recessive alleles is 25%. However, in X-linked inheritance, the sex of the child must also be considered.

Educational objective:

Although females inherit two copies of the X chromosome, males inherit only one. A consequence of this difference is that males cannot be carriers of an X-linked recessive allele. Therefore, the incidence of X-linked recessive conditions is usually much greater in males than in females.

Time spent:0 sec

Subject:

Biology

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System:

Genetics and Evolution